Problem 4 – Humanize

Abstract

We prove that the answer to the question is *no* in general. If (M,ω) is a symplectic manifold, $D \subset M$ a symplectic simple normal crossings divisor, and J an ω-tame almost complex structure admitting, locally on M, smooth complex coordinates $(z_1,\dots,z_n)$ with $z_1,\dots,z_r$ J-holomorphic and $D = \{z_1 \cdots z_r = 0\}$, then the normal bundle of every compact connected component D_α of D admits a flat structure with locally constant (locally) J-holomorphic trivialising frames; in particular

$$\langle e(\nu_{D_\alpha}), [D_\alpha] \rangle \;=\; D_\alpha \cdot D_\alpha \;=\; 0.$$

For a projective line $L \cong \mathbb{CP}^1$ in $(\mathbb{CP}^2, \omega_{\mathrm{FS}})$ (a symplectic SNC divisor with one component) one has $L \cdot L = 1$, so no such J exists, and the cohomologous deformation of ω allowed in the question cannot remove the obstruction, which is topological in the pair (M, D).

1 Statement and answer

Question. Let (M,ω) be a closed symplectic manifold and $D \subset M$ a symplectic simple normal crossings divisor. Does there exist an ω-tame almost complex structure J such that locally on M there are smooth complex coordinates $(z_1,\dots,z_n)$ in which, for some $0 \le r \le n$, the functions $z_1,\dots,z_r$ are J-holomorphic and D is given by $z_1 \cdots z_r = 0$? (One may first replace ω by a symplectic form connected to it through a path of symplectic forms, all cohomologous to ω, along which D remains a symplectic simple normal crossings divisor.)

Answer: No in general. The precise form is:

Theorem 1.1 (Obstruction)**.** *Let (M,ω) be a symplectic manifold, $D \subset M$ a symplectic simple normal crossings divisor, and let J be an almost complex structure on M which is* SNC-adapted *to D, meaning: every point $p \in M$ has a neighbourhood U with smooth complex coordinates $(z_1,\dots,z_n)\colon U \to U' \subset \mathbb{C}^n$ such that, for some $0 \le r \le n$,*

(i) $D \cap U = \{q \in U : z_1(q) \cdots z_r(q) = 0\}$*;*

(ii) *$z_1,\dots,z_r$ are J-holomorphic on U (the condition $\mathrm{d}z_k \circ J = i\,\mathrm{d}z_k$ holds at every point of U).*

Let D_α be a compact connected component of D. Then the normal bundle ν_α of D_α in M — a complex line bundle via the quotient of J, see Lemma 2.1 — possesses a flat connection; in particular its Euler class satisfies

$$e(\nu_\alpha) \;=\; 0 \;\in\; H^2(D_\alpha;\mathbb{R}), \qquad \textit{hence} \qquad \langle e(\nu_\alpha), [D_\alpha] \rangle \;=\; D_\alpha \cdot D_\alpha \;=\; 0.$$

(The conclusion of the question would therefore force $D_\alpha \cdot D_\alpha = 0$ for every compact component, regardless of the tameness of J — the latter enters the question but the obstruction needs no taming at all.)

Corollary 1.2 (The counterexample)**.** *Let $M = \mathbb{CP}^2$, $\omega = \omega_{\mathrm{FS}}$ the Fubini–Study symplectic form, and $D = L \cong \mathbb{CP}^1$ a projective line. Then D is a symplectic simple normal crossings divisor in the closed symplectic manifold (M, ω): it is a smooth symplectic submanifold, and a single component trivially constitutes simple normal crossings. But*

$$L \cdot L \;=\; \int_L c_1\big(\mathcal{O}_{\mathbb{CP}^1}(1)\big) \;=\; 1 \;\neq\; 0,$$

since $\nu_L \cong \mathcal{O}_{\mathbb{CP}^1}(1)$. By Theorem 1.1, no ω'-tame SNC-adapted J exists for any *symplectic form ω' in a cohomologous deformation family: the quantity $\langle e(\nu_L), [L]\rangle = L \cdot L$ is a topological intersection number of the pair (M, L), hence unchanged by any deformation of ω that keeps L smoothly embedded (and a fortiori symplectic SNC). Moreover, the same conclusion holds for the nodal SNC divisor $D = L_1 \cup L_2$ formed by two distinct lines: each component satisfies $L_i \cdot L_i = 1$.*

Remark 1.3 (What happens for components with trivial normal Euler class)*.* For components with $e(\nu_\alpha) = 0$ in $H^2(D_\alpha; \mathbb{R})$ (e.g. trivial normal bundle), Theorem 1.1 says nothing; and indeed in the product situation $M = X \times \mathbb{C}$ with $D = X \times \{0\}$ an ω-tame adapted J does exist whenever $(X, \omega|_X)$ carries a tame almost complex structure J_X: take $J = J_X \oplus i$. This shows the obstruction found here is specific to components with nontrivial normal Euler class, and the counterexample relies on $L \cdot L \neq 0$.

2 Strata are J-invariant

Lemma 2.1. *Let J be SNC-adapted to D (no taming assumption needed). Then for every $p \in D$, and $A := \{\alpha : p \in D_\alpha\}$, the tangent space T_pD_A of the stratum $D_A = \bigcap_{\alpha\in A} D_\alpha$ is a J_p-complex subspace of T_pM. Consequently $J_A := J|_{TD_A}$ is an almost complex structure on D_A, and, if J is ω-tame, so is J_A (with respect to $\omega|_{D_A}$).*

Proof. Choose adapted coordinates $(z_1, \dots, z_n)$ at p with $D \cap U = \{z_1 \cdots z_r = 0\}$, so that $D_A \cap U = \{z_k = 0 : \alpha_k \in A\}$ after numbering, and $T_pD_A = \bigcap_{k:\,\alpha_k\in A} \ker \mathrm{d}z_k(p)$. For $v \in \ker \mathrm{d}z_k(p)$, the J-holomorphicity of z_k gives $\mathrm{d}z_k(Jv) = i\,\mathrm{d}z_k(v) = 0$, so $Jv \in \ker \mathrm{d}z_k(p)$. Thus each $\ker \mathrm{d}z_k(p)$ is J_p-invariant, and so is their intersection. The statement about taming restricts from M to the invariant subspace: for $v \in T_pD_A \setminus 0$, $\omega(v, J_Av) = \omega(v, Jv) > 0$. □

Corollary 2.2 (Normal bundle is a complex line bundle)**.** *For any component D_α, the normal bundle $\nu_\alpha := TM|_{D_\alpha}/TD_\alpha$ carries a fibrewise complex structure induced by J, making it a complex line bundle over D_α; the conormal bundle $\nu_\alpha^* = \mathrm{Hom}_{\mathbb{C}}(\nu_\alpha, \mathbb{C})$ is the dual complex line bundle.*

3 The ratio lemma

Lemma 3.1 (Ratios are J-holomorphic across the divisor)**.** *Let J be SNC-adapted, and let ζ, ζ' be J-holomorphic functions on a connected open set $W \subset M$ with*

$$\{\zeta = 0\} \;=\; \{\zeta' = 0\} \;=\; W \cap D_\alpha, \qquad \mathrm{d}\zeta \neq 0,\ \mathrm{d}\zeta' \neq 0 \text{ on } W \cap D_\alpha.$$

Then the ratio $h := \zeta'/\zeta$, smooth and nonvanishing on $W \setminus D_\alpha$, extends to a smooth nonvanishing function on W, the extension satisfies

$$\mathrm{d}\zeta' \;=\; h\,\mathrm{d}\zeta \quad \text{on } W \cap D_\alpha, \tag{1}$$

and h is J-holomorphic on W: $\mathrm{d}h \circ J = i\,\mathrm{d}h$.

Proof. Off D_α, both functions are nowhere zero, so h is smooth nonvanishing there, and the computation $\mathrm{d}\zeta' = \mathrm{d}h\,\zeta + h\,\mathrm{d}\zeta$ composed with J gives, using $\mathrm{d}\zeta \circ J = i\mathrm{d}\zeta$ and $\mathrm{d}\zeta' \circ J = i\mathrm{d}\zeta'$,

$$0 \;=\; \zeta \cdot \big(\mathrm{d}h \circ J - i\,\mathrm{d}h\big),$$

hence $\mathrm{d}h \circ J = i\mathrm{d}h$ on $W \setminus D_\alpha$: h is J-holomorphic there.

For the extension: $D_\alpha \cap W$ is a smooth codimension-two submanifold along which both ζ, ζ' vanish with nonvanishing differential transversely. In any tubular neighbourhood with normal coordinate, both ζ and ζ' have a simple zero along the zero section; their quotient extends smoothly and nonvanishingly across the zero section (one-variable Taylor argument in the normal direction: if $f(t) = t\,F(t)$, $g(t) = t\,G(t)$ with $F(0), G(0) \neq 0$, then $f/g = F/G$ extends). Differentiating $\zeta' = h\zeta$ at $q \in D_\alpha$ (where $\zeta = 0$) yields (1), and since $\mathrm{d}\zeta'(q) \neq 0$ and $\mathrm{d}\zeta(q) \neq 0$, the extended h is nonvanishing also on D_α.

Finally, the identity $\mathrm{d}h \circ J = i\mathrm{d}h$ holds on $W \setminus D_\alpha$, both sides are continuous on W (since h and J are smooth on W), and $W \setminus D_\alpha$ is dense in W; hence the identity holds on all of W. □

4 The maximum principle

Lemma 4.1. *Let (X, J) be a connected almost complex manifold of dimension $2m \geq 2$, and $h\colon X \to \mathbb{C}$ J-holomorphic. If $|h|$ has a local maximum at some point of X, then h is constant. In particular, on a compact connected almost complex manifold every J-holomorphic function is constant.*

Proof. We use the local existence of J-holomorphic discs in almost complex manifolds (Nijenhuis–Woolf): for every $p \in X$ and every complex line $\ell \subset (T_pX, J_p)$ there is a J-holomorphic disc $\varphi\colon \Delta \to X$ with $\varphi(0) = p$ and $\varphi_* T_0\Delta = \ell$; moreover one can choose such discs smoothly depending on parameters (e.g. on the line and on nearby points), so that their images through $\{p'\} \times \{\text{lines in } T_{p'}X\}$ cover a neighbourhood of p.

Suppose $|h|$ attains a local maximum at p. For any small J-holomorphic disc $\varphi\colon \Delta \to X$ with $\varphi(0) = p$, the composition $h \circ \varphi$ is holomorphic $\Delta \to \mathbb{C}$ (chain rule from $\mathrm{d}h \circ J = i\mathrm{d}h$), and $|h \circ \varphi|$ attains a maximum at 0; by the classical maximum modulus principle $h \circ \varphi$ is constant, in particular $h|_{\varphi(\Delta)} \equiv h(p)$. Since such discs through p sweep a neighbourhood of p (choose the disc family as above), $h \equiv h(p)$ on an open set containing p. Thus $\{q : h(q) = h(p)\}$ is open; it is also closed, and X is connected, so $h \equiv h(p)$. The compact case follows because $|h|$ attains its maximum. □

5 Proof of the obstruction

Proof of Theorem 1.1. Let D_α be a compact connected component; all components are smooth codimension-2 symplectic submanifolds, and we use the conormal complex line bundle ν_α^* of Corollary 2.2.

Step 1: J-holomorphic conormal frames. By SNC-adaptedness, cover D_α by finitely many adapted coordinate neighbourhoods W_i with the property that $z_1^{(i)}$ is J-holomorphic and $\{z_1^{(i)} = 0\} = D_\alpha \cap W_i$: at any $p \in D_\alpha$ the component D_α is among the coordinate hyperplanes of every adapted coordinate system at p (it appears among the $D = \{z_1 \cdots z_r = 0\}$ factors), so renumbering puts it first; this works also at points of D_α lying on other components. On $W_i \cap D_\alpha$, set

$$e_i \;:=\; \mathrm{d}z_1^{(i)}\big|_{D_\alpha \cap W_i} \;\in\; \Gamma\big(D_\alpha \cap W_i,\; T^*M \otimes \mathbb{C}\big).$$

Since $z_1^{(i)}|_{D_\alpha \cap W_i} = 0$, e_i annihilates $T(D_\alpha)$, so it descends to a homomorphism $\nu_\alpha \to \mathbb{C}$; since $\mathrm{d}z_1^{(i)} \neq 0$ along D_α (it is a coordinate differential), $e_i \neq 0$ everywhere on $D_\alpha \cap W_i$; and by J-holomorphicity, $e_i(Jv) = \mathrm{d}z_1^{(i)}(Jv) = i\,\mathrm{d}z_1^{(i)}(v) = i\,e_i(v)$, so e_i is $\mathbb{C}$-linear for the complex structure of Corollary 2.2: e_i is a nowhere vanishing local frame of the complex line bundle ν_α^*.

Step 2: constant transition functions. On $W_i \cap W_j$, both $z_1^{(i)}$ and $z_1^{(j)}$ are J-holomorphic defining functions of $D_\alpha \cap W_i \cap W_j$, with the hypotheses of Lemma 3.1. Hence the ratio $h_{ij} := z_1^{(j)}/z_1^{(i)}$ extends smoothly and nonvanishingly across D_α, is J-holomorphic on $W_i \cap W_j$, and $e_j = h_{ij}\,e_i$ on $D_\alpha \cap W_i \cap W_j$ (this is (1)). Restricting to D_α: by Lemma 2.1, $J_\alpha := J|_{TD_\alpha}$ is an almost complex structure on the compact connected manifold D_α, and $h_{ij}|_{D_\alpha}$ is J_α-holomorphic: indeed $\mathrm{d}(h_{ij}|_{D_\alpha}) \circ J_\alpha = (\mathrm{d}h_{ij} \circ J)|_{TD_\alpha} = i\,\mathrm{d}h_{ij}|_{TD_\alpha} = i\,\mathrm{d}(h_{ij}|_{D_\alpha})$. By Lemma 4.1, $h_{ij}|_{D_\alpha}$ is constant on every connected component of $D_\alpha \cap W_i \cap W_j$.

Step 3: flatness and vanishing Euler class. The complex line bundle ν_α^* thus admits a trivialising cover (the e_i) whose transition functions $\{h_{ij}|_{D_\alpha}\}$ are constant on each connected overlap piece. Choose on each $W_i \cap D_\alpha$ the connection whose covariant derivative kills e_i; since on overlaps the frames differ by constants, these local connections agree on overlaps and assemble to a global connection ∇ on ν_α^* with $\nabla e_i = 0$ on W_i; its curvature satisfies $\Omega(e_i) = 0$ locally, hence $\Omega \equiv 0$: ∇ is flat. By Chern–Weil theory (the first Chern form of a flat connection is identically zero),

$$c_1(\nu_\alpha^*) \;=\; 0 \;\in\; H^2(D_\alpha;\mathbb{R}).$$

For a complex line bundle, the Euler class of the oriented real plane bundle equals c_1, and duals reverse the sign; hence

$$e(\nu_\alpha) \;=\; c_1(\nu_\alpha) \;=\; -c_1(\nu_\alpha^*) \;=\; 0, \qquad \langle e(\nu_\alpha), [D_\alpha]\rangle = 0.$$

The self-intersection $D_\alpha \cdot D_\alpha := \langle e(\nu_\alpha), [D_\alpha]\rangle$ therefore vanishes. □

Proof of Corollary 1.2. $(\mathbb{CP}^2, \omega_{\mathrm{FS}})$ is a closed symplectic manifold and a projective line L is a smooth symplectic hypersurface; SNC for a single component is automatic. Its normal bundle is $\mathcal{O}(1)$ with c_1 evaluating to 1 on $[L]$; so the conclusion of Theorem 1.1 fails: no ω_{FS}-tame SNC-adapted J exists.

For the deformation clause: suppose $\{\omega_t\}_{t\in[0,1]}$ is a path of cohomologous symplectic forms with $\omega_0 = \omega_{\mathrm{FS}}$, along which L remains a symplectic SNC divisor. Apply Theorem 1.1 to $(\mathbb{CP}^2, \omega_1, L)$: the hypothesis "$J$ is ω_1-tame and SNC-adapted" implies $L \cdot L = 0$; but $L \cdot L$ is the topological intersection number of the homology class $[L] \in H_2(\mathbb{CP}^2;\mathbb{Z})$ with itself, equal to 1 regardless of the symplectic form used (it only depends on the underlying oriented smooth pair $(\mathbb{CP}^2, L)$): contradiction. Hence no such J exists after any allowed deformation either. For $D = L_1 \cup L_2$ formed by two distinct lines: they meet transversely at one point, and the standard local Darboux model for two transversally intersecting symplectic hypersurfaces supplies the SNC charts at the intersection point, so D is a symplectic SNC divisor in the sense of the question; each component satisfies $L_i \cdot L_i = 1$, so the same conclusion holds. □

6 Discussion of sharpness

(1) *What the obstruction uses.* Only the existence of local J-holomorphic defining functions and the compactness of the component enter: the ratio lemma makes the conormal frames constant along the component via the maximum principle. Tameness of J, closedness of M, and the SNC

condition beyond smoothness of the component are superfluous for the obstruction — but $(\mathbb{CP}^2, L)$ satisfies all the hypotheses of the question, so the answer to the question as stated is no.
(2) *The deformation does not help because the obstruction is not symplectic.* Theorem 1.1 is a statement about J and D alone (given adaptedness); its conclusion $D_\alpha \cdot D_\alpha = 0$ is a topological fact of the pair. No change of ω within its cohomology class (or any class) affects it, as long as D_α stays compact.
(3) *Sharpness of the principle.* The obstruction vanishes when $D_\alpha \cdot D_\alpha = 0$; in the product situation $D = X \times \{0\} \subset X \times \mathbb{C}$ (or $X \times \Sigma^g$) with the product form, where ν_D is trivial, an adapted tame structure exists whenever X carries a tame almost complex structure J_X: take $J_X \oplus i$ (already noted after Corollary 1.2). Hence the obstruction used here cannot by itself distinguish such cases; whether the answer is yes whenever all components have vanishing normal Euler class is a further question not decided here.
(4) *More counterexamples.* Any closed symplectic (M, ω) containing a symplectic SNC divisor with a compact connected component D_α of nonzero self-intersection is a counterexample, stably under the allowed deformation; for instance a line in $\mathbb{CP}^2$ (self-intersection $+1$) or an exceptional sphere in a symplectic blow-up (self-intersection -1). By contrast, a smooth fibre of a ruled surface has self-intersection 0, so the present obstruction does not apply to it.